

Japan's Toyota Builds a City-like Center to Test Robotics, AI

Japanese automaker Toyota says it is building a research center in a city-like setting to test robotics, artificial intelligence (AI) and autonomous vehicle technologies.

Toyota recently announced it had completed the first part, or phase, of the center, called Woven City. It sits near the southern Japanese city of Susono. Company officials recently showed off the latest progress to reporters from the Associated Press.

Daisuke Toyoda is an executive for Toyota who is helping lead the project. He told the AP that the center was not designed to be a “smart city.” However, it aims to carry out technology research and development and be “a test course for mobility.”

The company said Woven City would serve as a “Living Laboratory.” It aims to test technology systems to support future changes in “the movement of people, goods, information and energy.”

Toyota executives in charge of Woven City, from left to right, Woven by Toyota CFO Kenta Kon, CEO Hajime Kumabe and Head of Woven City Management Daisuke Toyoda speak to reporters in Susono, Japan Feb. 22, 2025. (AP Photo/Yuri Kageyama)

Toyoda said the center is meant to be a place where researchers and technology company officials can come together and share ideas. It will also seek to establish a community with a shared desire “to co-create, develop and refine” new technology products and services.

Woven City was built on the grounds of a closed Toyota automobile factory. The automaker said the project’s first phase covers about 47,000 square meters. When completed, it will spread out over 294,000 square meters.

Building operations on Woven City began in 2021. All the buildings are connected by underground passageways. Among planned testing activities will include self-driving vehicles making waste pickups and completing deliveries around the area. Testing operations will center on how people living in cities can best interact with changing technologies.

Currently, no one lives in Woven City. When it opens, officials plan on having about 100 people living there. They will be called “weavers.” These are workers employed by Toyota and its partner companies making other products.

This photo provided by Woven by Toyota shows the square at the center of the apartment complexes of Woven City in Susono city, Shizuoka Prefecture on Feb. 2025. (Woven by Toyota via AP)

When AP reporters visited the area, Japanese coffee maker UCC was serving hot drinks from a self-driving bus. The bus was parked in an area surrounded by still-empty housing.

Toyota has supported electric vehicle (EV) technology in the past. However, the company is currently involved in a push for hydrogen, the energy of choice in Woven City.

Keisuke Konishi is an automobile expert at Japan's Quick Corporate Valuation Research Center. He told the AP Toyota has plans to expand into self-driving vehicle services to compete with Google's Waymo and other large companies.

This photo provided by Woven by Toyota shows the sign of Toyota Woven City in Susono city, Shizuoka Prefecture on Feb. 2025. (Woven by Toyota via AP)

Konishi noted the company has the money for such development even if it means building up a completely new business. "Toyota has the money to do all that," he said.

Toyota officials have said they do not expect Woven City to make money, at least not for the first few years.

Several other futuristic developments have been planned in other areas of the world over the years. They have included efforts in Toronto, Canada; Saudi Arabia; Abu Dhabi; and San Francisco, California. But those projects are either still being developed or have been canceled.

I'm Bryan Lynn.

The Associated Press reported this story. Bryan Lynn adapted the report for VOA Learning English.

Vocabulary

autonomous - adj. อัตโนมัติ. สามารถทำงานได้โดยไม่ต้องอาศัยความช่วยเหลือจากมนุษย์

mobility - n. การเคลื่อนที่. ความสามารถในการเคลื่อนย้ายหรือเดินทางไปมาได้อย่างอิสระ

refine - v. กลั่นกรอง/ปรับปรุง. ทำให้บางสิ่งบริสุทธิ์หรือปรับปรุงบางสิ่งให้ดีขึ้น โดยเฉพาะอย่างยิ่งโดยการขจัดวัสดุที่ไม่ต้องการออกไป

deliver - v. ส่งมอบ. เคลื่อนย้ายสินค้าจากที่หนึ่งไปยังอีกที่หนึ่ง

park - v. จอด. นำยานพาหนะไปไว้ในที่ที่สามารถจอดพักอยู่ได้เป็นระยะเวลาหนึ่ง

Reading Comprehension Quiz

1. What is the primary purpose of the research center Toyota is building?

- A) To manufacture new Toyota vehicles for global markets.
- B) To test robotics, AI, and autonomous vehicle technologies.

- C) To serve as a new headquarters for Toyota's executives.
- D) To train future employees in traditional car manufacturing.

2. Where is Woven City primarily located?

- A) In the heart of Tokyo, Japan.
- B) Near the southern Japanese city of Susono.
- C) On an island off the coast of Japan.
- D) In a suburb of Nagoya, Japan.

3. According to Daisuke Toyoda, what is Woven City *not* designed to be?

- A) A test course for mobility.
- B) A center for technology research and development.
- C) A smart city.
- D) A place for collaboration.

4. What alternative name does the company use to describe Woven City's function?

- A) A Commercial Hub.
- B) A Living Laboratory.
- C) An Entertainment Complex.
- D) A Historical Museum.

5. Besides technology research, what social objective does Woven City aim to achieve regarding its community?

- A) To create a closed community exclusive to Toyota employees.
- B) To establish a community with a shared desire to co-create, develop, and refine new technology.
- C) To provide affordable housing for the general public.
- D) To host international conferences on urban planning.

6. Woven City was built on the grounds of what type of former facility?

- A) A disused power plant.
- B) A closed Toyota automobile factory.
- C) An abandoned amusement park.
- D) A former agricultural research station.

7. When did building operations for Woven City begin?

- A) 2018

- B) 2020
- C) 2021
- D) 2023

8. What specific examples of self-driving vehicle activities are planned for testing in Woven City?

- A) Public transportation and emergency services.
- B) Waste pickups and completing deliveries.
- C) Construction site supervision and material transport.
- D) Personal taxi services and tourist tours.

9. What is the current status of Woven City regarding residents?

- A) It is fully populated with 100 residents.
- B) It has no residents currently, but plans to host 100 people.
- C) It houses only research scientists from partner companies.
- D) It is open to the public for short-term visits.

10. Who are the "weavers" expected to be living in Woven City?

- A) Local artists and craftspeople.
- B) Retirees from Toyota's former factory.
- C) Workers employed by Toyota and its partner companies.
- D) Government officials overseeing the project.

11. During the AP reporters' visit, what unique service was observed in Woven City?

- A) A self-driving public bus offering free rides.
- B) A drone delivering packages to residents.
- C) A self-driving bus serving hot drinks.
- D) Robotic cleaners maintaining public spaces.

12. What is the primary energy source of choice for Woven City, according to the article?

- A) Solar power.
- B) Electric vehicle (EV) technology.
- C) Hydrogen.
- D) Natural gas.

13. According to automobile expert Keisuke Konishi, what is Toyota's strategic goal regarding self-driving vehicle services?

- A) To solely focus on manufacturing self-driving hardware.
- B) To expand into self-driving vehicle services to compete with companies like Google's Waymo.
- C) To license its self-driving technology to other automakers.
- D) To only develop self-driving vehicles for internal company use.

14. What is Keisuke Konishi's assessment of Toyota's financial capacity for new business development like Woven City?

- A) He believes Toyota lacks the necessary funds.
- B) He is unsure about Toyota's financial stability.
- C) He states Toyota has the money to do all that.
- D) He suggests Toyota needs external investment.

15. How does the article characterize other futuristic city developments mentioned around the world?

- A) Most have been fully completed and are operational.
- B) They are all experiencing significant financial success.
- C) They are either still being developed or have been canceled.
- D) They primarily focus on residential housing rather than technology.